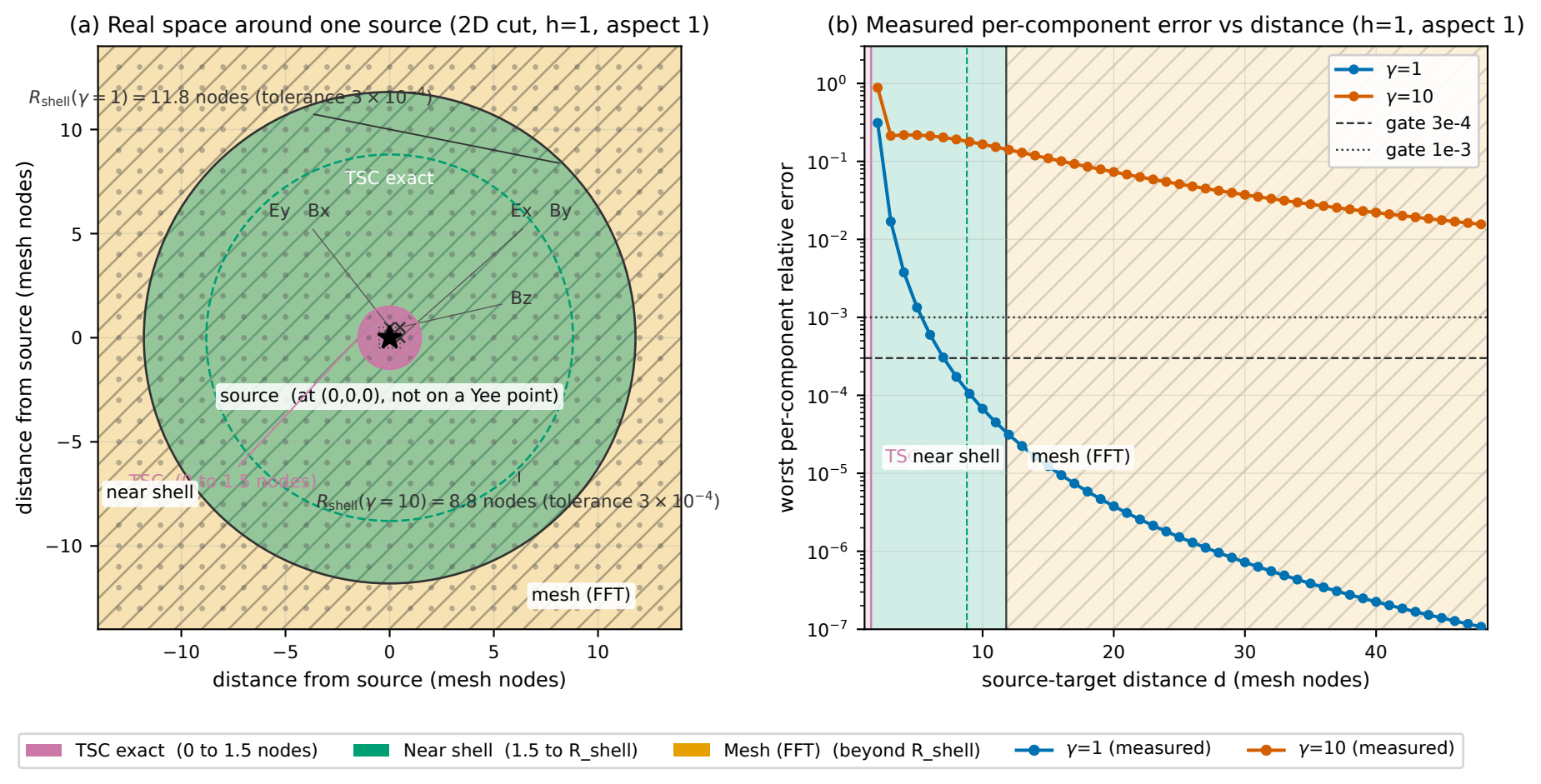

(a) Real space around one source (2D cut, h=1, aspect 1)
$R_{shell}(\gamma = 1) = 11.8$ nodes (tolerance 3×10^{-4})
TSC exact
Ey Bx
Ex By
Bz
source (at (0,0,0), not on a Yee point)
TSC (0 to 1.5 nodes)
near shell
$R_{shell}(\gamma = 10) = 8.8$ nodes (tolerance 3×10^{-4})
mesh (FFT)
distance from source (mesh nodes)
distance from source (mesh nodes)
(b) Measured per-component error vs distance (h=1, aspect 1)
γ=1
γ=10
gate 3e-4
gate 1e-3
TSC
near shell
mesh (FFT)
worst per-component relative error
source-target distance d (mesh nodes)
TSC exact (0 to 1.5 nodes)
Near shell (1.5 to R_shell)
Mesh (FFT) (beyond R_shell)
γ=1 (measured)
γ=10 (measured)